

Harnessing LLMs for Medical Insights: NER Extraction from Summarized Medical Text

Problem Statement: The paper talks about to produce concise, coherent summaries of medical literature, prioritizing clinical relevance. Automated NER is adopted to extract vital entities from the summaries generated by LLMs.

Algorithms Used: More research is undertaken toward NER techniques. PEGASUS LLM from Google – MLM(Encoder-Masked Language Model) and GSG(Decoder-Gap Sentence Generation), T5 (transformer based encoder-decoder model for supervised and self-supervised learning) and GPT-3.

Datasets: PubMed Database managed by NCBI. Accessible via E-utilities, that supports XML and JSON format for searching and retrieving records.

Model Training and Testing: Models are trained for 10 epochs. *ROUGE i.e. Recall-Oriented Understudy for Gisting Evaluation* and BERT metrics has been used to evaluate the summaries produced by the models.

Results: Models performance comparison for metrics evaluation is as follows

Model Name	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore_Precision	BERTScore_Recall	BERTScore_F1score
T5	0.4008	0.1350	0.2426	0.6314	0.6240	0.6256
Pegasus	0.4402	0.1742	0.2506	0.6378	0.6610	0.6468
GPT3.5	0.4573	0.1783	0.2784	0.6565	0.6617	0.6585

Conclusions: This study presented large language models and named entity recognition to summarize and extract information from the vast PubMed medical literature database. State-of-the-art LLMs like T5 were fine-tuned to generate concise summaries, while a BERT-based NER model identified critical biomedical entities within the summaries.

Relevance to Our Team: NER techniques used could be applied in the model. Performance of the model could be improvised in pre-processing.

Biomedical Chat Assistant with Personalized Document Reader Using BioMistral and RAG

Problem Statement: This paper reports the development of a Biomedical Chat Assistant equipped with a personalized reader for documents, using the capabilities provided by the BioMistral framework and Retrieval-Augmented Generation techniques.

Algorithms Used: BioMistral LLM integrated with RAG. BioMistral supports in understanding biomedical text. Document chunking pubmedbert.

Datasets: PDF documents on clinical research

Results: Biomedical Chat Assistant by analyzing the various established methods such as PubMed Search API, BioBERT, Clinical BERT, and Semantic Scholar.

Metric	Biomedical Chat Assistant	PubMed Search API	BioBERT	Clinical BERT	Semantic Scholar
Accuracy	92%	75%	80%	78%	76%
Response Time	1.5 seconds	4 seconds	3 seconds	3.5 seconds	5 seconds
Precision	88%	70%	75%	72%	73%
Recall	85%	60%	70%	65%	68%
F1 Score	0.86	0.67	0.77	0.72	0.71
User Satisfaction Rating	4.5/5	3.5/5	3.8/5	3.6/5	3.7/5
User Engagement	7 interactions/session	3-4 interactions/session	3-4 interactions/session	3-4 interactions/session	3-4 interactions/session

Conclusions: The system efficiently processed numerous unstructured PDF medical reports to retrieve relevant information with high accuracy.

Open Challenges: Further refinements are needed to address challenges like document format variability and medical language complexity.

Relevance to Our Team: BioMistral LLM and RAG, Chroma vector to store the Vector embeddings and Recursive Text Splitter for document chunking can be relevant to our project.

The Impact of LoRA Adapters on LLMs for Clinical Text Classification Under Computational and Data Constraints

Problem Statement: Fine-tuning Large Language Models (LLMs) for clinical Natural Language Processing (NLP) poses significant challenges due to domain gap, limited data, and stringent hardware constraints. In this paper, the four adapter techniques—Adapter, Lightweight, TinyAttention, and Gated Residual Network (GRN)- equivalent to Low-Rank Adaptation (LoRA), for clinical note classification under real world, resource-constrained conditions are evaluated.

Algorithms Used: Fine tuning of pre-trained LLM models(CamemBERT-bio, AliBERT, DrBERT) with LoRA, freezing the base model incur

1. Huge computational power
2. Expensive and slow
3. Need large datasets

Two light weight Transformer models are fine tuned using adapters outperform adapter augmented LLMs that are trained from scratch. This proved more efficient for the low in-resource, limited data for training and compute.

Datasets: Clinical notes dataset of France

Results: Accuracy is more when using GRN adapter for the Transformer models.

Performance comparison of transformer with different adapters.

Transformer	Acc (↑)	Pre (↑)	Rec (↑)	F1 (↑)	Training Time (hours) (↓)	Inference Time (s) (↓)
Baseline	0.85	0.85	0.83	0.84	0.11	3
Adapter	0.85	0.83	0.85	0.84	0.4	2
Lightweight	0.85	0.82	0.88	0.85	1	3
GRN	0.87	0.85	0.89	0.87	0.7	3
TinyAttention	0.85	0.81	0.88	0.84	0.7	3

Bold denotes the best values.

TABLE 5. Performance comparison of AliBERT with different adapters.

AliBERT	Acc (↑)	Pre (↑)	Rec (↑)	F1 (↑)	Training Time (hours) (↓)	Inference Time (s) (↓)
Setup 1						
Baseline	0.86	0.87	0.84	0.86	39.3	128
Adapter	0.78	0.72	0.88	0.79	42.8	128
Lightweight	0.84	0.82	0.85	0.84	34.8	131
GRN	0.87	0.84	0.84	0.85	30.8	127
TinyAttention	0.84	0.81	0.83	0.82	42.2	128
Setup 2						
Adapter	0.68	0.67	0.67	0.67	46.6	126
Lightweight	0.66	0.72	0.49	0.58	46.7	126
GRN	0.67	0.7	0.59	0.64	37.5	126
TinyAttention	0.67	0.67	0.7	0.68	46.2	130

Bold denotes the best values.

TABLE 6. Performance comparison of DrBERT with different adapters.

DrBERT	Acc (↑)	Pre (↑)	Rec (↑)	F1 (↑)	Training Time (hours) (↓)	Inference Time (s) (↓)
Setup 1						
Baseline	0.87	0.84	0.9	0.87	45.2	133
Adapter	0.86	0.87	0.87	0.87	38.5	126
Lightweight	0.71	0.78	0.56	0.65	41.6	132
GRN	0.86	0.84	0.88	0.86	41.3	130
TinyAttention	0.85	0.81	0.9	0.85	28.9	123
Setup 2						
Adapter	0.69	0.71	0.63	0.67	46.1	122
Lightweight	0.73	0.72	0.73	0.72	45	125
GRN	0.73	0.7	0.76	0.73	47	126
TinyAttention	0.75	0.76	0.69	0.72	47.5	123

Bold denotes the best values.

Conclusions: Transformer-based models trained from scratch perform comparably or better, especially in environments with limited computational resources and data availability. Among the adapter structures evaluated, the GRN demonstrated superior accuracy, precision, recall, and F1 score, making it the most effective adapter for enhancing clinical note classification.

Open Challenges:

1. With the use singleGPU, we lacked the memory capacity and compute throughput required to fine-tune large proprietary models like Deepseek and Grok
2. Compared fully trainable versus fully frozen backbones with adapters; intermediate freezing ratios (e.g., 40-80% of layers frozen) may affect convergence speed and generalization, but were not explored.
3. Performance analysis focused primarily on accuracy and training time; other metrics such as memory footprint, latency, and energy consumption in diverse hospital server environments were not measured.

Relevance to Our Team: Hardware resource and the memory matters for precise computation during Query Answering in medical data.